RATIONALITY CRITERIA

Abstract. Let $X := (X_1, \ldots, X_d)$ be a tuple in a tracial W^*-probability space. We establish three criteria for affiliated operators to be rational in X, each with several equivalent characterizations. The first criterion is of algebraic nature and assumes only $\Delta(X) = d$ for X. From it, we derive a second criterion of an analytic nature using liberation dual systems. When X consists of normal generators and admits a dual system, we obtain a third criterion that further refines the second in terms of finite-rank fraction commutators, thereby extending the rationality criteria of Duchamp and Reutenauer, Linnell, and Miyagawa for affiliated operators.

1. Introduction

Evaluating a noncommutative rational function at a tuple X of bounded operators can yield an unbounded affiliated operator. The free-field realization theorem of Mai, Speicher and Yin [MSY23, Theorem 1.1] shows that evaluation at X embeds the free field into the $*$-algebra of affiliated operators when $\Delta(X) = d$. Under this condition, we first establish algebraic characterizations of rationality in X. We then derive two analytic criteria: one using liberation dual systems and the other using dual systems.

Our question is motivated by Connes's rationality conjecture for free group algebras [Con94, Section IV.5, Remark 3], a noncommutative analogue of Kronecker's criterion [Kro81]. In its operator formulation, this criterion characterizes rationality by a finite-rank commutator condition [Con94, Section IV.3, Proposition 1]. For a free group, Connes observed that commutators with the operator arising from its Cayley tree have finite rank on the rational closure of the group algebra in the reduced group C^*-algebra, and conjectured the converse.

Duchamp and Reutenauer [DR97, Theorem 7] proved this conjecture for a free group $\mathbb{F}_d$ with d generators ($d \geq 1$), and showed that the rational closure coincides with the division closure. Their proof uses noncommutative rational series. Using the bounded dual system for free Haar unitaries (see, for example, [MSY23, Proposition 6.3]), one can express the Cayley-tree finite-rank condition as finite rank of the corresponding commutators.

Linnell [Lin00, Theorem 1.1] extends the free-group criterion to affiliated operators using left and right fraction representations. Theorem 1.9 extends this criterion to normal generators admitting a dual system in the sense of Definition 1.8, including free Haar unitaries with their bounded dual systems. Linnell also identifies the bounded part of the affiliated division closure with the division closure in the reduced group C^*-algebra. Together with this identity, Theorem 1.9 recovers the criterion of Duchamp and Reutenauer.

Miyagawa [Miy22, Theorems 3.1 and 4.5] proves a semicircular analogue of these criteria, characterizing rationality by finite-rank commutators with the right annihilation operators (whose negatives form a bounded dual system in our convention). His proof adapts the noncommutative rational-series approach of Duchamp and Reutenauer to the semicircular setting. He also extends this characterization to affiliated operators using left and right fraction representations.

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1.1. **A rationality criterion under maximal Δ.** Our first main result gives several equivalent characterizations under maximal Δ alone. The quantity Δ was introduced by Connes and Shlyakhtenko in their paper [CS05] on the homology of von Neumann algebras.

Throughout the paper, $(\mathcal{M},\tau)$ is a tracial W^*-probability space, that is, a von Neumann algebra $\mathcal{M}$ equipped with a faithful normal tracial state τ. Let $X=(X_1,\dots,X_d)\in\mathcal{M}^d$, with $d\geq 1$. Put $H:=L^2(\mathcal{M},\tau)$ and $\Omega:=1$. Inner products on H are linear in the second variable, and we identify elements of $\mathcal{M}$ with left multiplication on H. The condition $\Delta(X)=d$ means that there are no nonzero finite-rank operators $T_i\in\mathcal{F}(H)$ such that

$$\sum_i [\rho(X_i),T_i]=0;$$

see Section 2.5 for details. Here $\rho(X_i)$ denotes the right multiplication operator by X_i on H.

We write $L^0(\mathcal{M})$ for the $*$-algebra of closed densely defined operators affiliated with $\mathcal{M}$. We identify H with the square-integrable affiliated operators in $L^0(\mathcal{M})$ [AP18, Theorem 7.3.2]. In particular, $H\otimes_{\mathbb{C}}H\subseteq L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}$. Write $\mathcal{A}_X:=\mathbb{C}\langle X_1,\dots,X_d\rangle$ for the algebra generated by $X_1,\dots,X_d$. We denote its division closure in $L^0(\mathcal{M})$ by $\mathrm{Div}_{L^0(\mathcal{M})}\mathcal{A}_X$ and call an affiliated operator *rational* when it belongs to this closure; see Definition 2.1.

Definition 1.1. For $u\in L^0(\mathcal{M})$, a *fraction pair* is a quadruple $Q=(s,a,b,t)\in\mathcal{M}^4$ satisfying

$$u=s^{-1}a=bt^{-1},\qquad s,t \text{ invertible in } L^0(\mathcal{M}). \tag{1}$$

Definition 1.2. Let D be a linear operator on H with domain $\mathrm{Dom}\,D$. For a fraction pair $Q=(s,a,b,t)$ of an element $u\in L^0(\mathcal{M})$, define the *fraction commutator of D and Q* by

$$\Gamma[Q,D]:=sDb-aDt, \tag{2}$$

with domain given by $\{\xi\in H: b\xi,t\xi\in\mathrm{Dom}\,D\}$.

The identities $ut=b$ and $su=a$ from (1) give formally

$$s[D,u]t=sD(ut)-(su)Dt=sDb-aDt=\Gamma[Q,D].$$

That is, s and t remove the fractions in u from the commutator expression.

Multiplication in $L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}$ is given by

$$(u_1\otimes v_1^{\mathrm{op}})(u_2\otimes v_2^{\mathrm{op}}):=u_1u_2\otimes(v_2v_1)^{\mathrm{op}}.$$

To state our first main result, we use the right multiplication maps in [CS05, Section 3.1.1, equation (3.1)], with the coefficient algebra enlarged from $\mathcal{M}\otimes_{\mathbb{C}}\mathcal{M}^{\mathrm{op}}$ to $L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}$.

Definition 1.3. For $w\in L^0(\mathcal{M})$, define the right multiplication map

$$\Psi_w: L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}\longrightarrow L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}$$

by

$$\Psi_w(k):=k(w\otimes 1-1\otimes w^{\mathrm{op}}) \tag{3}$$

For $u,v\in L^0(\mathcal{M})$, this gives

$$\Psi_w(u\otimes v^{\mathrm{op}})=uw\otimes v^{\mathrm{op}}-u\otimes(wv)^{\mathrm{op}}.$$

For a tuple $X=(X_1,\dots,X_d)\in\mathcal{M}^d$, we further write $\Psi_i:=\Psi_{X_i}$.

Theorem 1.4. *Suppose $\Delta(X)=d$. For $u\in L^0(\mathcal{M})$ the following are equivalent:*

1. $u\in\mathrm{Div}_{L^0(\mathcal{M})}\mathcal{A}_X$.
2. *There are $k_i\in L^0(\mathcal{M})\otimes_{\mathbb{C}}L^0(\mathcal{M})^{\mathrm{op}}$ such that*

$$\sum_{i=1}^d\Psi_i(k_i)=u\otimes 1-1\otimes u^{\mathrm{op}}. \tag{4}$$

(3) *For every fraction pair* $Q = (s, a, b, t)$ *for* u*, there exist* $\ell_1, \dots, \ell_d \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ *such that*

$$\sum_{i=1}^{d} \Psi_i(\ell_i) = a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}}. \tag{5}$$

(4) *There exist a fraction pair* $Q = (s, a, b, t)$ *for* u *and tensors* $\ell_1, \dots, \ell_d \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}$ *satisfying* (5)*.*

(5) *There exist a fraction pair* $Q = (s, a, b, t)$ *for* u *and a tuple* $T = (T_1, \dots, T_d) \in \mathcal{F}(H)^d$ *such that*

$$\sum_{i=1}^{d} [\rho(X_i), T_i] = \Gamma[Q, P_\Omega]. \tag{6}$$

In (2)*,* $k_1, \dots, k_d$ *are uniquely determined by* u*. The tensors* $\ell_1, \dots, \ell_d$ *in* (3) *and* (4) *and the tuple* T *in* (5) *are uniquely determined by the fraction pair* Q*.*

Here P_Ω denotes the orthogonal projection of H onto $\mathbb{C}\Omega$. By Definition 1.2,

$$\Gamma[Q, P_\Omega] = sP_\Omega b - aP_\Omega t \in \mathcal{F}(H).$$

The left-hand sides of equations (4) and (6) are values of degree-one Hochschild differentials with the corresponding coefficient bimodules; see Remark 3.4.

Remark 1.5. When the equivalent assertions of Theorem 1.4 hold, the tensors in assertion (2) are the free difference quotients $k_i = \partial_i u$ of u with respect to $X_1, \dots, X_d$; see the implication (1) $\implies$ (2) in its proof. The implication (2) $\implies$ (3) follows as Ψ_i is a right multiplication map. The implication (3) $\implies$ (4) is obtained by clearing denominators in ℓ_i. The implication (4) $\implies$ (5) is due to the identification of the tensors with finite-rank operators. The key input for (5) $\implies$ (1) is that $\mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$ is a division ring, as guaranteed under $\Delta(X) = d$ by [MSY23, Theorem 1.1].

1.2. **Analytic criteria.** Condition (5) in Theorem 1.4 requires a fraction pair Q for which the commutator equation (6) has the (unique) finite-rank solution $T \in \mathcal{F}(H)^d$. The next theorem gives a rationality criterion that is valid for every fraction pair. That is, (6) admits a finite-rank solution either for every fraction pair or for none.

To formulate this analytic criterion, we write $\mathcal{M}\Omega := \{x\Omega : x \in \mathcal{M}\} \subset H$. We use the following variant of the (liberation) dual systems in [Voi99, Definition 13.1].

Definition 1.6. Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$. A *liberation dual system* for X is a tuple $(L_1, \dots, L_d)$ of linear operators on H such that

$$\mathcal{M}\Omega \subseteq \mathrm{Dom}\, L_i \cap \mathrm{Dom}\, L_i^* \qquad (1 \le i \le d),$$

and

$$[X_j, L_i]\xi = \delta_{ij}[X_i, P_\Omega]\xi \qquad (\xi \in \mathcal{M}\Omega,\ 1 \le i, j \le d).$$

We call this system *normalized* if

$$\sum_{i=1}^{d} L_i \xi = P_\Omega \xi \qquad (\xi \in \mathcal{M}\Omega). \tag{7}$$

Every liberation dual system can be normalized, as shown in Lemma 4.1.

Theorem 1.7. *Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$, and suppose that $X_i - \lambda$ is invertible in $L^0(\mathcal{M})$ for every $1 \le i \le d$ and every $\lambda \in \mathbb{C}$. Let $(L_1, \dots, L_d)$ be a normalized liberation dual system for X in the sense of Definition 1.6. Then, for every $u \in L^0(\mathcal{M})$, the following are equivalent:*

(1) $u \in \mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$.
(2) *There exist a fraction pair Q for u and $T_1, \dots, T_d \in \mathcal{F}(H)$ such that*

$$\sum_{i=1}^{d} [\rho(X_i), T_i] = \Gamma[Q, P_\Omega]. \tag{8}$$

(3) *For every fraction pair Q for u, there exist $T_1, \dots, T_d \in \mathcal{F}(H)$ satisfying (8).*
(4) *For every fraction pair Q for u and every i,*

$$\overline{\Gamma[Q, L_i]} \in [\rho(X_i), \mathcal{F}(H)].$$

For each fixed fraction pair Q, the tuple $(T_1, \dots, T_d)$ in assertions (2) and (3) is unique.

Here $[\rho(X_i), \mathcal{F}(H)] := \{[\rho(X_i), T] : T \in \mathcal{F}(H)\}$ and $\overline{\Gamma[Q, L_i]}$ denotes the closure of the closable operator $\Gamma[Q, L_i]$. The new characterization in assertion (4) can be understood as follows. For any fraction pair Q, assertion (4) provides $T_i \in \mathcal{F}(H)$ with $[\rho(X_i), T_i] = \overline{\Gamma[Q, L_i]}$. Summing over i and using the normalization in equation (7), we obtain

$$\sum_i [\rho(X_i), T_i] = \sum_i \overline{\Gamma[Q, L_i]} = \Gamma[Q, P_\Omega],$$

so $(T_1, \dots, T_d)$ is the solution in assertion (3).

Theorems 1.4 and 1.7 do not require $X_1, \dots, X_d$ to be normal. For normal generators, we obtain two rationality criteria. In Section 6, we apply Theorem 1.7 to normal generators with nonatomic spectral distributions and L^2 liberation gradients. The case of generators admitting conjugate variables follows as a corollary.

Our third main result further treats normal generators admitting a dual system and identifies the finite-rank solutions T_i with the closures of the corresponding fraction commutators. We use the following variant of the dual systems in [Voi98, Definition 5.1].

Definition 1.8. Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$. A *dual system* for X is a tuple $(D_1, \dots, D_d)$ of linear operators on H such that

$$\mathcal{M}\Omega \subseteq \mathrm{Dom}\, D_i \cap \mathrm{Dom}\, D_i^*, \qquad [X_j, D_i]|_{\mathcal{M}\Omega} = \delta_{ij} P_\Omega|_{\mathcal{M}\Omega} \qquad (1 \le i, j \le d). \tag{9}$$

Theorem 1.9. *Suppose that $X_1, \dots, X_d \in \mathcal{M}$ are normal operators generating $\mathcal{M}$, and let $(D_1, \dots, D_d)$ be a dual system for X in the sense of Definition 1.8. For every $u \in L^0(\mathcal{M})$ and every fraction pair $Q = (s, a, b, t)$ for u, the following are equivalent:*

(1) $u \in \mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$.
(2) *For every i, $\Gamma[Q, D_i]$ has finite rank.*
(3) *For every i, $\overline{\Gamma[Q, D_i]} \in \mathcal{F}(H)$.*

When these equivalent conditions hold, the rank of each $\overline{\Gamma[Q, D_i]}$ is independent of Q.

Theorem 1.9 identifies the finite-rank operators in Theorem 1.7(4) with fraction-commutator closures for a dual system. Lemma 5.6 constructs a normalized liberation dual system $(L_1, \dots, L_d)$ from $(D_1, \dots, D_d)$ and verifies regularity of every scalar shift. For this system,

$$\overline{\Gamma[Q, L_i]} = [\rho(X_i), \overline{\Gamma[Q, D_i]}]$$

whenever $\overline{\Gamma[Q, D_i]}$ is bounded. Consequently, when these closures have finite rank, they are the finite-rank solution required in Theorem 1.7(2), i.e.,

$$\sum_{i=1}^{d} [\rho(X_i), \overline{\Gamma[Q, D_i]}] = \Gamma[Q, P_\Omega].$$

Conversely, normalization and Lemma 5.3 identify the finite-rank operators required in Theorem 1.7 (4) with these closures.

A dual system for normal generators yields conjugate variables; the arcsine tuple of Remark 6.8 has a normalized liberation dual system but no conjugate variable, hence no dual system. The arcsine tuple therefore satisfies the hypotheses of Theorem 1.7 but not those of Theorem 1.9. Hence the hypotheses of Theorem 1.9 are strictly stronger than those of Theorem 1.7.

At present, we do not know whether normal generators admitting conjugate variables necessarily admit a dual system in the sense of Definition 1.8. We present the two criteria in this order for convenience of exposition.

For normal operators admitting bounded dual systems, the fraction commutators are bounded and $\overline{\Gamma[Q, D_i]} = \Gamma[Q, D_i]$. Theorem 1.9 therefore recovers the criteria of Duchamp and Reutenauer and Linnell for free Haar unitaries, and of Miyagawa for free semicircular operators.

1.3. **Organization.** Section 2 collects the algebraic preliminaries on division closures and the free field, followed by fractions of affiliated operators, finite-rank tensor realizations and the consequences of maximal Δ. Section 3 proves Theorem 1.4 using denominator clearing and the tensor realization of finite-rank operators. Section 4 proves that liberation dual systems give maximal Δ when every $X_i - \lambda$ is regular, and establishes Theorem 1.7 by recovering algebraic L^2 tensors. Section 5 proves independence of the fraction pair, identifies finite-rank solutions with fraction-commutator closures, and constructs a normalized liberation dual system to deduce Theorem 1.9 from Theorem 1.7(4). Section 6 constructs normalized liberation dual systems from L^2 liberation gradients and applies Theorem 1.7 to normal generators with nonatomic spectral distributions. It then obtains the conjugate-variable case as Corollary 6.7 and gives an arcsine example separating the two hypotheses. Section 7 uses a circular operator to illustrate the role of normality in Theorem 1.9.

2. Preliminaries

We first recall division closures, the free field and an algebraic tensor criterion for membership in a division subring. We then discuss fractions of affiliated operators, realize algebraic tensors as finite-rank operators and recall the consequences of maximal Δ.

2.1. Division closures and the free field.

Definition 2.1. For a unital subalgebra A of a unital algebra B, its *division closure* $\mathrm{Div}_B A$ is the smallest unital subalgebra containing A and closed under inversion of its elements that are invertible in B; see [Coh85, Exercise 7.1.4] and [MSY23, Definition 4.10]. An affiliated operator u is *rational in* X if $u \in \mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$.

The related notion of *rational closure* is defined using entries of matrix inverses [MSY23, Definition 4.7]; see also [Coh71, Section 7]. It contains the division closure and may be strictly larger in general [Coh85, Exercise 7.1.4]. Since the division and rational closures coincide when $\Delta(X) = d$ (see Subsection 2.5), we use *rational in* X for membership in the division closure.

The free field $\mathbb{C}\langle\langle x_1, \ldots, x_d \rangle\rangle$ is the universal division ring of fractions of the free polynomial algebra $\mathbb{C}\langle x_1, \ldots, x_d \rangle$; see [MSY23, Section 4.1 and Theorem 4.1] and [Coh06, Corollary 7.5.14].

It is the universal localization of this algebra at all full square matrices [Coh06, Theorem 7.5.13 and Corollary 7.5.14].

The free difference quotients on noncommutative polynomials [Voi98, Section 3, pp. 192–193] extend uniquely to the free field [BD78, Section 2.1, (4), pp. 298–299]. An explicit construction of these extensions on noncommutative rational functions is given in [KVV12, Section 4.4, Definition 4.4 and Corollary 4.11]. For $1 \le i \le d$, we use the linear identification $u \otimes v \mapsto u \otimes v^{\mathrm{op}}$ to write the extended maps as

$$\partial_i : \mathbb{C}\langle\langle x_1, \ldots, x_d \rangle\rangle \longrightarrow \mathbb{C}\langle\langle x_1, \ldots, x_d \rangle\rangle \otimes_{\mathbb{C}} \mathbb{C}\langle\langle x_1, \ldots, x_d \rangle\rangle^{\mathrm{op}}.$$

For $c \in \mathbb{C}$ and $u, v \in \mathbb{C}\langle\langle x_1, \ldots, x_d \rangle\rangle$, with $u \neq 0$, they satisfy

$$\begin{aligned} \partial_i c = 0, \qquad & \partial_i x_j = \delta_{ij} 1 \otimes 1, \\ \partial_i(uv) = (1 \otimes v^{\mathrm{op}})\partial_i u + (u \otimes 1)\partial_i v, \quad & \partial_i(u^{-1}) = -(u^{-1} \otimes (u^{-1})^{\mathrm{op}})\partial_i u. \end{aligned} \tag{10}$$

2.2. **Coordinate maps over division subrings.** For a division subring F in a unital algebra U, the proof below constructs, for $u \notin F$, a right F-linear coordinate map $\phi_u : U \longrightarrow F$ satisfying $\phi_u(1) = 0$ and $\phi_u(u) = 1$. This map distinguishes $u \otimes_F 1$ from $1 \otimes_F u$, which is recorded as the lemma below. This is the key lemma used in the proof of Theorem 1.4 to force an affiliated operator to be a rational function in X.

Lemma 2.2. *Let U be a nonzero unital algebra and let $F \subseteq U$ be a division subring containing the identity of U. Then, for every $u \in U$,*

$$u \otimes_F 1 = 1 \otimes_F u \text{ in } U \otimes_F U \quad \Longleftrightarrow \quad u \in F.$$

Proof. If $u \in F$, the identity follows from the balancing relation

$$(v_1 a) \otimes_F v_2 = v_1 \otimes_F (a v_2), \qquad v_1, v_2 \in U, \quad a \in F.$$

If $u \notin F$, the elements $1, u$ are right F-linearly independent: $v_1 + u v_2 = 0$ with $v_2 \neq 0$ would give $u = -v_1 v_2^{-1} \in F$. Extend (1,u) to a basis of the right F-module U by [Hun74, Theorem IV.2.4]. The coordinate map ϕ_u at u is right F-linear (see [Hun74, Theorem IV.2.1 & Theorem 4.11]) and satisfies $\phi_u(1) = 0$, $\phi_u(u) = 1$. The map $(v_1, v_2) \mapsto \phi_u(v_1)v_2$ is middle linear and hence induces an additive map [Hun74, Definition IV.5.1 and Theorem IV.5.2]

$$U \otimes_F U \longrightarrow U, \qquad v_1 \otimes_F v_2 \longmapsto \phi_u(v_1) v_2$$

which sends $u \otimes_F 1 - 1 \otimes_F u$ to 1, so this tensor cannot vanish. □

2.3. **Fractions of affiliated operators.** We now turn to fraction representations in $L^0(\mathcal{M})$. Affiliation means commutation with each unitary in the commutant, including invariance of the domain; see [AP18, Section 7.2] and [MSY23, Section 5.1].

Definition 2.3. An element $s \in \mathcal{M}$ is *regular* if it is invertible in $L^0(\mathcal{M})$.

Regularity is equivalent to injectivity and dense range on H [MSY23, Lemmas 5.1–5.2].

For $u \in L^0(\mathcal{M})$, a fraction pair is given by

$$s := (1 + uu^*)^{-1/2}, \quad a := su, \qquad t := (1 + u^*u)^{-1/2}, \quad b := ut, \tag{11}$$

where $s, a, b, t \in \mathcal{M}$, s, t are regular, and $u = s^{-1}a = bt^{-1}$. These facts follow from functional calculus and polar decomposition for affiliated operators [MSY23, Section 5.1].

Definition 2.4. For a fraction pair $Q = (s, a, b, t)$ of u as in Definition 1.1, the representations $s^{-1}a$ and bt^{-1} are called a *left fraction* and a *right fraction*, respectively; see [Lüc02, Section 8.2.1].

For $u_1, \ldots, u_m \in L^0(\mathcal{M})$, a regular $s \in \mathcal{M}$ is a *common left denominator* if $su_j \in \mathcal{M}$ for every j; a regular $t \in \mathcal{M}$ is a *common right denominator* if $u_j t \in \mathcal{M}$ for every j; see [Lin00, Section 1] and [Miy22, Lemma 4.1].

Common denominators for finite families are used in [Lin00, Section 1, p. 2 of the arXiv version]. The following proof iterates the two-element construction of [Miy22, Lemma 4.1].

Lemma 2.5. *For finite families $u_1,\dots,u_m$ and $v_1,\dots,v_n$ in $L^0(\mathcal{M})$ there are regular $s,t\in\mathcal{M}$ such that*

$$su_j\in\mathcal{M}\quad(1\le j\le m),\qquad v_kt\in\mathcal{M}\quad(1\le k\le n).$$

Proof. Choose a regular $s\in\mathcal{M}$ with $su_1\in\mathcal{M}$ using (11). Suppose $su_j\in\mathcal{M}$ for $1\le j\le\ell$, where $1\le\ell<m$. Using (11), write

$$u_{\ell+1}=s_1^{-1}a_1,\qquad ss_1^{-1}=s_2^{-1}a_2,$$

where $s_1,s_2,a_1,a_2\in\mathcal{M}$ and s_1,s_2 are regular. Then $s':=s_2s=a_2s_1$ is regular, and

$$s'u_j=s_2(su_j)\in\mathcal{M}\quad(1\le j\le\ell),\qquad s'u_{\ell+1}=a_2a_1\in\mathcal{M}.$$

This completes the induction for the left denominators. The proof for right denominators is similar. □

Lemma 2.6 ([Miy22, proof of Lemma 4.1])**.** *If $s_1,s_2\in\mathcal{M}$ are regular, they have a common left multiple $xs_1=ys_2$ with $x,y\in\mathcal{M}$ regular. If $t_1,t_2\in\mathcal{M}$ are regular, they have a common right multiple $t_1x'=t_2y'$ with $x',y'\in\mathcal{M}$ regular. Both common multiples are regular.*

Proof. Choose a left fraction $s_1s_2^{-1}=(s')^{-1}a$ using (11). Thus we may take $x:=s'$ and $y:=a$. Indeed, $s's_1=as_2$, and $a=s's_1s_2^{-1}$ is invertible in $L^0(\mathcal{M})$, hence regular. A right fraction $t_1^{-1}t_2=b_3t_3^{-1}$ gives the second identity, with $b_3=t_1^{-1}t_2t_3$ regular. Products of regular elements are regular. □

Linnell [Lin00, Theorem 1.1] characterizes rational affiliated operators over a free group by finite-rank conditions on bounded operators obtained from left and right fractions. Miyagawa [Miy22, Section 4] gives a semicircular analogue using the expressions

$$sr_i^*b-ar_i^*t=\Gamma[Q,r_i^*],$$

where $Q=(s,a,b,t)$ is a fraction pair and r_i^* are the bounded right annihilation operators.

2.4. **Algebraic tensors and finite-rank operators.** Write $\mathcal{F}(H)$ for the finite-rank bounded operators on H. For $x\in\mathcal{M}$, let $\rho(x)\in\mathcal{B}(H)$ denote the unique bounded extension of right multiplication

$$\rho(x)(y\Omega):=(yx)\Omega,\qquad y\in\mathcal{M}.$$

We use the commutator convention $[S,T]:=ST-TS$.

Let $S_2(H)$ be the Schatten class of Hilbert–Schmidt operators, that is, the Hilbert space of operators $S\in\mathcal{B}(H)$ with $\operatorname{Tr}(S^*S)<\infty$, equipped with the Hilbert–Schmidt norm $\|S\|_{\mathrm{HS}}:=\operatorname{Tr}(S^*S)^{1/2}$. Here Tr is the usual trace on $\mathcal{B}(H)$ [AP18, Section 8.4].

Let J be Tomita's conjugation operator on H, given by $J(x\Omega):=x^*\Omega$; see [AP18, Section 7.1.1] and [MSY23, p. 13]. It satisfies $J^2=I$ and $\rho(x)=Jx^*J$. Following [CS05, Section 1.1, equation (1.1)], define

$$\Theta(u\otimes v)\xi:=u\langle Jv,\xi\rangle,\qquad\Theta:H\otimes_{\mathbb{C}}H\longrightarrow\mathcal{F}(H).\tag{12}$$

Here $u,v\in H=L^2(\mathcal{M},\tau)\subset L^0(\mathcal{M})$ and $\xi\in H$. The map Θ is a linear bijection and extends to a unitary

$$\Theta:H\bar{\otimes}H\longrightarrow S_2(H).$$

The actions in [CS05, equation (1.2)] give, for $a,b\in\mathcal{M}$ and $u,v\in H$,

$$\Theta(au\otimes vb)=a\Theta(u\otimes v)b.$$

Via the linear identification $u \otimes v \mapsto u \otimes v^{\mathrm{op}}$, we also regard Θ as a map on the corresponding subspace of $L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$. For $u, v \in \mathcal{M}$, equation (12) reads

$$\Theta(u \otimes v^{\mathrm{op}}) = uP_\Omega v. \tag{13}$$

For $x \in \mathcal{M}$ and $u, v \in H$, the right multiplication operator $\rho(x)$ satisfies

$$\rho(x)\Theta(u \otimes v^{\mathrm{op}}) = \Theta(ux \otimes v^{\mathrm{op}}),$$
$$\Theta(u \otimes v^{\mathrm{op}})\rho(x) = \Theta(u \otimes (xv)^{\mathrm{op}}).$$

Subtracting gives

$$[\rho(x), \Theta(u \otimes v^{\mathrm{op}})] = \Theta(ux \otimes v^{\mathrm{op}} - u \otimes (xv)^{\mathrm{op}}). \tag{14}$$

We extend τ to H by $\tau(\xi) := \langle \Omega, \xi \rangle$.

2.5. The invariant Δ and free-field realization.

Definition 2.7. The von Neumann tensor product $\mathcal{M} \bar{\otimes} \mathcal{M}^{\mathrm{op}}$, where $\mathcal{M}^{\mathrm{op}}$ is the opposite algebra, acts on $S_2(H)$ by

$$(a \otimes b^{\mathrm{op}}) \cdot S := aSb, \qquad a, b \in \mathcal{M}.$$

For $X = (X_1, \dots, X_d) \in \mathcal{M}^d$, set

$$\mathcal{C}_X := \left\{ T = (T_1, \dots, T_d) \in \mathcal{F}(H)^d : \sum_{i=1}^{d} [\rho(X_i), T_i] = 0 \right\}.$$

We call an element $T \in \mathcal{C}_X$ a *finite-rank cycle*. Such tuples represent Hochschild 1-cycles with Hilbert–Schmidt coefficients; see [CS05]. Following [CS05] and [MSY23, Section 3], define

$$\Delta(X) := d - \dim_{\mathcal{M} \bar{\otimes} \mathcal{M}^{\mathrm{op}}} \overline{\mathcal{C}_X}^{\mathrm{HS}}. \tag{15}$$

The closure is taken in $S_2(H)^d$. The dimension is the von Neumann dimension [Lüc02, Definition 1.10] for the product trace $(a \otimes b^{\mathrm{op}}) \mapsto \tau(a)\tau(b)$; see also [CS05, Definition 2.10]. Thus $\Delta(X) \le d$.

We recall the finite-rank characterization of maximal Δ and the resulting free-field realization of the division closure, both from [MSY23].

Remark 2.8. By [MSY23, Section 3 and Remark 3.2], for every $X \in \mathcal{M}^d$,

$$\Delta(X) = d \iff \mathcal{C}_X = \{0\} \iff \left\{ T \in \mathcal{F}(H)^d : \sum_i [T_i, X_i] = 0 \right\} = \{0\}.$$

Lemma 2.9 ([MSY23, Theorems 1.1, 3.1 and 5.6])**.** *If $\Delta(X) = d$, evaluation at X extends to an embedding of the free field $\mathbb{C}\langle\langle x_1, \dots, x_d \rangle\rangle$ into $L^0(\mathcal{M})$. Its image is the division subring* $\mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$.

If $\Delta(X) = d$, the rational closure and division closure of $\mathcal{A}_X$ in $L^0(\mathcal{M})$ coincide [MSY23, Theorem 1.1 and Remark 5.10].

3. Rationality under maximal Δ

We prove Theorem 1.4 in this section. By Definition 1.3, Ψ_i is right multiplication by $X_i \otimes 1 - 1 \otimes X_i^{\mathrm{op}}$. In particular,

$$\Psi_i(rk) = r\Psi_i(k), \qquad r, k \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}. \tag{16}$$

Lemma 3.1. *Given $k_1, \dots, k_d \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$, there are regular $s, t \in \mathcal{M}$ such that*

$$(s \otimes t^{\mathrm{op}})k_i \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}} \qquad (1 \le i \le d).$$

Proof. For simplicity of notation, we give the calculation for simple tensors

$$k_i = u_i \otimes v_i^{\mathrm{op}}, \qquad u_i, v_i \in L^0(\mathcal{M}).$$

The same argument applies to general finite sums. Applying Lemma 2.5 to $u_1, \dots, u_d$ gives a regular $s \in \mathcal{M}$ such that

$$su_i \in \mathcal{M} \qquad (1 \le i \le d).$$

Applying its right-denominator assertion to $v_1, \dots, v_d$ gives a regular $t \in \mathcal{M}$ such that

$$v_i t \in \mathcal{M} \qquad (1 \le i \le d).$$

Then

$$(s \otimes t^{\mathrm{op}})k_i = su_i \otimes (v_i t)^{\mathrm{op}} \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}.$$

□

Lemma 3.2. *Suppose $\Delta(X) = d$. The map*

$$(L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}})^d \longrightarrow L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}, \qquad (k_i) \longmapsto \sum_i \Psi_i(k_i)$$

is injective.

Proof. Suppose $\sum_i \Psi_i(k_i) = 0$. Choose regular $s, t \in \mathcal{M}$ as in Lemma 3.1 and set $T_i := \Theta((s \otimes t^{\mathrm{op}})k_i) \in \mathcal{F}(H)$. By (14) and (16),

$$\sum_i [\rho(X_i), T_i] = \Theta\left((s \otimes t^{\mathrm{op}}) \sum_i \Psi_i(k_i)\right) = 0.$$

Remark 2.8 gives $T_i = 0$ for every i. Injectivity of Θ implies $(s \otimes t^{\mathrm{op}})k_i = 0$. Left multiplication by $s^{-1} \otimes (t^{-1})^{\mathrm{op}}$ then gives $k_i = 0$. □

Lemma 3.3. *Let $Q = (s, a, b, t)$ be a fraction pair for $u \in L^0(\mathcal{M})$, and let $\ell_1, \dots, \ell_d \in H \otimes_{\mathbb{C}} H$. Set $T_i := -\Theta(\ell_i) \in \mathcal{F}(H)$. Then*

$$\sum_i \Psi_i(\ell_i) = a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}}$$

if and only if

$$\sum_i [\rho(X_i), T_i] = \Gamma[Q, P_\Omega].$$

Proof. By (14) and (13),

$$\sum_i [\rho(X_i), T_i] = -\Theta\left(\sum_i \Psi_i(\ell_i)\right), \qquad \Gamma[Q, P_\Omega] = -\Theta(a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}}).$$

The equivalence follows from the injectivity of Θ. □

By Lemma 2.9, $\mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$ is a division subring of $L^0(\mathcal{M})$ whenever $\Delta(X) = d$. We prove Theorem 1.4 through (1) $\Longrightarrow$ (2) $\Longrightarrow$ (3) $\Longrightarrow$ (4) $\Longrightarrow$ (5) $\Longrightarrow$ (1).

Proof of Theorem 1.4. In this proof, set $U := L^0(\mathcal{M})$ and $F := \mathrm{Div}_U \mathcal{A}_X$.

(1) $\Longrightarrow$ (2). By [MSY23, Theorem 1.1, (i) and (v)], the assumption $\Delta(X) = d$ implies that $X_1, \dots, X_d$ generate the free field: evaluation identifies $\mathbb{C}\langle\langle x_1, \dots, x_d\rangle\rangle$ with F. The free difference quotients from Subsection 2.1, transported through this evaluation isomorphism, define linear maps

$$\partial_i : F \longrightarrow F \otimes_{\mathbb{C}} F^{\mathrm{op}}.$$

They satisfy the rules in equation (10) with x_j replaced by X_j. Put $\delta v := v \otimes 1 - 1 \otimes v^{\mathrm{op}}$. Direct expansion gives

$$\delta(v_1 v_2) = (1 \otimes v_2^{\mathrm{op}})\delta v_1 + (v_1 \otimes 1)\delta v_2, \qquad \delta(v^{-1}) = -(v^{-1} \otimes (v^{-1})^{\mathrm{op}})\delta v.$$

For constants, $\delta c = 0$, while for generators, $\delta X_j = \Psi_j(1 \otimes 1) = \sum_i \Psi_i(\partial_i X_j)$. We now show that this identity extends from constants and generators to all rational elements using the product and inverse rules.

If $\delta v_\ell = \sum_i \Psi_i(\partial_i v_\ell)$ for $\ell = 1, 2$, the product rules and equation (16) give

$$\begin{aligned}\delta(v_1 v_2) &= (1 \otimes v_2^{\mathrm{op}}) \sum_i \Psi_i(\partial_i v_1) + (v_1 \otimes 1) \sum_i \Psi_i(\partial_i v_2) \\ &= \sum_i \Psi_i((1 \otimes v_2^{\mathrm{op}})\partial_i v_1 + (v_1 \otimes 1)\partial_i v_2) \\ &= \sum_i \Psi_i(\partial_i(v_1 v_2)).\end{aligned}$$

Similarly, if $0 \neq v \in F$ satisfies $\delta v = \sum_i \Psi_i(\partial_i v)$, the inverse rules give

$$\begin{aligned}\delta(v^{-1}) &= -(v^{-1} \otimes (v^{-1})^{\mathrm{op}}) \sum_i \Psi_i(\partial_i v) \\ &= \sum_i \Psi_i(-(v^{-1} \otimes (v^{-1})^{\mathrm{op}})\partial_i v) \\ &= \sum_i \Psi_i(\partial_i(v^{-1})).\end{aligned}$$

Together with linearity, this shows that the elements satisfying the identity form a division subalgebra of F containing $\mathbb{C}$ and $X_1, \ldots, X_d$. Since these generate F, the identity holds for every rational element $u \in F$. This proves equation (4) with $k_i := \partial_i u$; thus assertion (2) holds. Lemma 3.2 proves uniqueness of the tensors in assertion (2).

(2) $\Longrightarrow$ (3). Given tensors k_i as in assertion (2), for any fraction pair $Q = (s, a, b, t)$ set $\ell_i := (s \otimes t^{\mathrm{op}})k_i$. By equation (16),

$$\sum_i \Psi_i(\ell_i) = (s \otimes t^{\mathrm{op}})(u \otimes 1 - 1 \otimes u^{\mathrm{op}}) = a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}},$$

so assertion (3) holds.

(3) $\Longrightarrow$ (4). Choose a fraction pair $Q = (s, a, b, t)$ for u and corresponding tensors ℓ_i in assertion (3). By Lemma 3.1, there are regular $s', t' \in \mathcal{M}$ such that

$$(s' \otimes (t')^{\mathrm{op}})\ell_i \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}.$$

By equation (16) and equation (5),

$$\sum_i \Psi_i((s' \otimes (t')^{\mathrm{op}})\ell_i) = s'a \otimes (tt')^{\mathrm{op}} - s's \otimes (bt')^{\mathrm{op}}.$$

Since $(s's, s'a, bt', tt')$ is a fraction pair for u, assertion (4) holds.

For fixed Q, uniqueness in assertions (3) and (4) follows from Lemma 3.2, applied to the difference of two solutions of equation (5).

(4) $\Longrightarrow$ (5). Let Q and ℓ_i be as in assertion (4). Lemma 3.3 gives the finite-rank operators $T_i := -\Theta(\ell_i)$ satisfying equation (6). Thus assertion (5) holds.

(5) $\Longrightarrow$ (1). Suppose assertion (5) holds. Let $Q = (s, a, b, t)$ be the fraction pair and $T_1, \ldots, T_d \in \mathcal{F}(H)$ the operators satisfying equation (6). Via Θ, define a linear map $\pi : \mathcal{F}(H) \longrightarrow U \otimes_F U$ by

$$\pi(\Theta(v_1 \otimes v_2^{\mathrm{op}})) := v_1 \otimes_F v_2, \qquad v_1, v_2 \in H.$$

Note that Θ is a linear bijection, so π is well defined. For a rank-one operator $\Theta(v_1 \otimes v_2^{\mathrm{op}})$, equation (14) gives

$$\begin{aligned}\pi([\rho(X_i), \Theta(v_1 \otimes v_2^{\mathrm{op}})]) &= (v_1 X_i) \otimes_F v_2 - v_1 \otimes_F (X_i v_2) \\ &= 0.\end{aligned}$$

The last equality uses $X_i \in F$ and the balancing relation

$$(uf) \otimes_F w = u \otimes_F (fw), \qquad u, w \in U, \quad f \in F.$$

Every finite-rank operator is a finite sum of rank-one operators, so linearity gives $\pi([\rho(X_i), T]) = 0$ for every $T \in \mathcal{F}(H)$.

Applying π to equation (6) and using $\Theta(v_1 \otimes v_2^{\mathrm{op}}) = v_1 P_\Omega v_2$ for $v_1, v_2 \in \mathcal{M}$ (see equation (13)), we obtain

$$\begin{aligned}0 &= \pi\left(\sum_i [\rho(X_i), T_i]\right) \\ &= \pi(\Gamma[Q, P_\Omega]) \\ &= \pi(sP_\Omega b - aP_\Omega t) \\ &= \pi\big(\Theta(s \otimes b^{\mathrm{op}}) - \Theta(a \otimes t^{\mathrm{op}})\big) \\ &= s \otimes_F b - a \otimes_F t.\end{aligned}$$

Hence $a \otimes_F t = s \otimes_F b$.

To clear the denominators, consider the map

$$\Phi_{s,t} : U \otimes_F U \longrightarrow U \otimes_F U, \qquad \Phi_{s,t}(u \otimes_F w) := s^{-1}u \otimes_F wt^{-1}.$$

It is well defined because it preserves the balancing relation. Applying $\Phi_{s,t}$ to $a \otimes_F t = s \otimes_F b$ yields

$$s^{-1}a \otimes_F tt^{-1} = s^{-1}s \otimes_F bt^{-1}.$$

Since $u = s^{-1}a = bt^{-1}$, this is

$$u \otimes_F 1 = 1 \otimes_F u.$$

Lemma 2.2 therefore gives $u \in F = \mathrm{Div}_U \mathcal{A}_X$, so assertion (1) holds.

For fixed Q, let $(T_1, \ldots, T_d)$ and $(S_1, \ldots, S_d)$ be two finite-rank solutions of equation (6). Their right-hand sides are both $\Gamma[Q, P_\Omega]$, so subtraction gives

$$\sum_i [\rho(X_i), T_i - S_i] = \sum_i [\rho(X_i), T_i] - \sum_i [\rho(X_i), S_i] = 0.$$

Each $T_i - S_i$ has finite rank. Since $\Delta(X) = d$, Remark 2.8 implies $T_i - S_i = 0$ for every i. Hence the two tuples coincide, proving uniqueness of the finite-rank solution tuple for the fixed fraction pair Q.

□

Remark 3.4. Equip $L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ with the $L^0(\mathcal{M})$-bimodule structure defined, for $u, v \in L^0(\mathcal{M})$ and $k \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$, by

$$u \cdot k \cdot v := k(v \otimes u^{\mathrm{op}}).$$

The degree-one Hochschild differential with these coefficients [Wei94, Section 9.1.1, p. 300] is the map

$$d_1 : (L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}) \otimes_{\mathbb{C}} L^0(\mathcal{M}) \longrightarrow L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$$

given by

$$d_1(k \otimes u) := k \cdot u - u \cdot k = k(u \otimes 1 - 1 \otimes u^{\mathrm{op}}).$$

Put $V_X := \operatorname{span}_{\mathbb{C}}\{X_1, \dots, X_d\}$. Restricting the last tensor factor to V_X, we obtain

$$\partial_X : (L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}) \otimes_{\mathbb{C}} V_X \longrightarrow L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}, \qquad \partial_X := d_1|_{(L^0(\mathcal{M})\otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}})\otimes_{\mathbb{C}} V_X}.$$

By Definition 1.3,

$$\partial_X\left(\sum_i k_i \otimes X_i\right) = \sum_i \Psi_i(k_i).$$

This restriction to the linear span of the generators follows [CS05, Section 3.1.1, equation (3.1)], with $L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ as the coefficient bimodule. Thus condition (2) in Theorem 1.4 requires $u \otimes 1 - 1 \otimes u^{\mathrm{op}}$ to lie in the image of ∂_X.

Similarly, equip $\mathcal{F}(H)$ with the $\mathcal{A}_X$-bimodule structure

$$x \cdot T \cdot y := \rho(y) T \rho(x), \qquad x, y \in \mathcal{A}_X, \quad T \in \mathcal{F}(H).$$

The degree-one Hochschild differential is then

$$d_1 : \mathcal{F}(H) \otimes_{\mathbb{C}} \mathcal{A}_X \longrightarrow \mathcal{F}(H), \qquad d_1(T \otimes x) := T \cdot x - x \cdot T = [\rho(x), T].$$

Since $\mathcal{A}_X$ is generated by $X_1, \dots, X_d$, the identity

$$[\rho(xy), T] = [\rho(y), \rho(x)T] + [\rho(x), T\rho(y)]$$

shows, by induction on monomials, that restricting the last tensor factor to V_X gives the full image of d_1:

$$\operatorname{im} d_1 = d_1(\mathcal{F}(H) \otimes_{\mathbb{C}} V_X) = \left\{ \sum_{i=1}^{d} [\rho(X_i), T_i] : T_i \in \mathcal{F}(H) \right\}.$$

Condition (6) is therefore

$$d_1\left(\sum_i T_i \otimes X_i\right) = \Gamma[Q, P_\Omega].$$

Thus $\Gamma[Q, P_\Omega]$ is a boundary, or equivalently its class vanishes in $H_0(\mathcal{A}_X, \mathcal{F}(H)) := \mathcal{F}(H)/\operatorname{im} d_1$.

4. Liberation dual system criterion

We prove Theorem 1.7 in this section.

Lemma 4.1. *If $X \in \mathcal{M}^d$ admits a liberation dual system, then it admits a normalized liberation dual system.*

Proof. Let $(L_1, \dots, L_d)$ be a liberation dual system for X. Define operators $C, \widetilde{L}_1, \dots, \widetilde{L}_d$, each with domain $\mathcal{M}\Omega$, by

$$C\xi := \sum_{k=1}^{d} L_k \xi - P_\Omega \xi, \qquad \widetilde{L}_i \xi := L_i \xi - \frac{1}{d} C\xi \qquad (\xi \in \mathcal{M}\Omega,\ 1 \le i \le d).$$

Since $X_j \mathcal{M}\Omega \subseteq \mathcal{M}\Omega$, the liberation relations give, on $\mathcal{M}\Omega$,

$$[X_j, C] = \sum_{k=1}^{d} [X_j, L_k] - [X_j, P_\Omega] = \sum_{k=1}^{d} \delta_{kj} [X_k, P_\Omega] - [X_j, P_\Omega] = 0.$$

Consequently, for $\xi \in \mathcal{M}\Omega$ and $1 \le i, j \le d$,

$$[X_j, \widetilde{L}_i]\xi = [X_j, L_i]\xi = \delta_{ij}[X_i, P_\Omega]\xi, \qquad \sum_{i=1}^{d} \widetilde{L}_i \xi = \sum_{i=1}^{d} L_i \xi - C\xi = P_\Omega \xi.$$

Since $\mathcal{M}\Omega \subseteq \operatorname{dom} L_i^*$, we have for $\eta, \xi \in \mathcal{M}\Omega$,

$$\langle \eta, \widetilde{L}_i \xi \rangle = \left\langle L_i^* \eta - \frac{1}{d} \sum_{k=1}^{d} L_k^* \eta + \frac{1}{d} P_\Omega \eta, \xi \right\rangle.$$

By the definition of the adjoint, $\eta \in \operatorname{dom} \widetilde{L}_i^*$, and hence $\mathcal{M}\Omega \subseteq \operatorname{dom} \widetilde{L}_i^*$. Thus $\widetilde{L}_1, \ldots, \widetilde{L}_d$ satisfy all the defining conditions. □

Lemma 4.2 (see [Kat95, Problem III.5.18])**.** *Let T be a densely defined closable operator on a Hilbert space H. If T has finite rank, then $\overline{T} \in \mathcal{F}(H)$, with the same range and rank as T.*

Lemma 4.3. *Let $x \in \mathcal{M}$ and suppose that $x - \lambda$ is regular for every $\lambda \in \mathbb{C}$. The map Ψ_x of Definition 1.3 is injective. If $k \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ satisfies*

$$\Psi_x(k) \in H \otimes_{\mathbb{C}} H,$$

then $k \in H \otimes_{\mathbb{C}} H$.

Proof. If $T \in \mathcal{F}(H)$ satisfies $[T, x] = 0$, then $\operatorname{ran} T$ is a finite-dimensional invariant subspace for x. A nonzero such subspace would contain an eigenvector, contradicting regularity of $x - \lambda$. Thus $T = 0$, so Remark 2.8 gives $\Delta(x) = 1$. Lemma 3.2, applied to the one-element tuple (x), now shows that Ψ_x is injective.

To prove the remaining assertion, put $U := L^0(\mathcal{M})$ and define the right multiplication maps

$$R_x : U \longrightarrow U, \qquad R_x(u) := ux,$$
$$R_{x^{\mathrm{op}}} : U^{\mathrm{op}} \longrightarrow U^{\mathrm{op}}, \quad R_{x^{\mathrm{op}}}(v^{\mathrm{op}}) := (xv)^{\mathrm{op}}.$$

Thus $\Psi_x = R_x \otimes I - I \otimes R_{x^{\mathrm{op}}}$. Neither right multiplication map has a nonzero finite-dimensional invariant subspace: such a subspace would contain an eigenvector, contradicting regularity of $x - \lambda$. Since $x \in \mathcal{M}$, both maps preserve H, and their restrictions to H have the same property.

For linear maps $A : V \to V$ and $B : W \to W$ on complex vector spaces, $A \otimes I - I \otimes B$ is injective on $V \otimes_{\mathbb{C}} W$ if B has no nonzero finite-dimensional invariant subspace. Indeed, write a nonzero kernel vector as $\sum_{r=1}^{n} v_r \otimes w_r$ with n minimal, so both families are linearly independent. Projecting the second factor modulo $\operatorname{span}\{w_1, \ldots, w_n\}$ shows that this span is B-invariant, a contradiction. The same conclusion holds with A in place of B.

The induced linear maps on the quotient spaces are

$$\widetilde{R}_x : U/H \longrightarrow U/H, \qquad \widetilde{R}_x(u + H) := ux + H,$$
$$\widetilde{R}_{x^{\mathrm{op}}} : U^{\mathrm{op}}/H \longrightarrow U^{\mathrm{op}}/H, \quad \widetilde{R}_{x^{\mathrm{op}}}(v^{\mathrm{op}} + H) := (xv)^{\mathrm{op}} + H.$$

Projecting the first factor of k modulo H gives a vector in the kernel of

$$\widetilde{R}_x \otimes I - I \otimes R_{x^{\mathrm{op}}} \quad \text{on } (U/H) \otimes_{\mathbb{C}} U^{\mathrm{op}}.$$

This map is injective by the preceding observation, so exactness of tensoring over $\mathbb{C}$ gives $k \in H \otimes_{\mathbb{C}} U^{\mathrm{op}}$. Projecting the second factor modulo H now gives a vector in the kernel of

$$(R_x|_H) \otimes I - I \otimes \widetilde{R}_{x^{\mathrm{op}}} \quad \text{on } H \otimes_{\mathbb{C}} (U^{\mathrm{op}}/H).$$

This map is injective because $R_x|_H$ has no nonzero finite-dimensional invariant subspace. It follows that $k \in H \otimes_{\mathbb{C}} H$. □

For a liberation dual system $(L_1, \ldots, L_d)$, define linear maps

$$\Phi_i : \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}} \longrightarrow \operatorname{Hom}_{\mathbb{C}}(\mathcal{M}\Omega, H), \qquad \Phi_i(c \otimes d^{\mathrm{op}}) := cL_i d|_{\mathcal{M}\Omega}.$$

The liberation commutator relations give, for $z \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}$ and $s, t \in \mathcal{M}$,

$$\Phi_i(\Psi_j(z)) = \delta_{ij} \Theta(\Psi_i(z)), \qquad \Phi_i((s \otimes t^{\mathrm{op}})z) = s\Phi_i(z)t \quad \text{on } \mathcal{M}\Omega. \tag{17}$$

Indeed, the first identity on $c \otimes d^{\mathrm{op}}$ is $c[X_j, L_i]d = \delta_{ij} c[X_i, P_\Omega]d$, and the second follows from multiplication in the opposite factor.

Proposition 4.4. *Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$, and suppose that $X_i - \lambda$ is regular for every $1 \le i \le d$ and every $\lambda \in \mathbb{C}$. If X admits a liberation dual system, then $\Delta(X) = d$.*

Proof. Let $T_1, \dots, T_d \in \mathcal{F}(H)$ satisfy $\sum_i [\rho(X_i), T_i] = 0$, and put $k_i := \Theta^{-1}(T_i)$. Equation (14) gives $\sum_i \Psi_i(k_i) = 0$. By Lemma 3.1, choose regular $s, t \in \mathcal{M}$ such that $m_i := (s \otimes t^{\mathrm{op}})k_i \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}$ for every i. Equation (16) yields $\sum_i \Psi_i(m_i) = 0$. Applying Φ_i and equation (17), then using density of $\mathcal{M}\Omega$, we obtain

$$\Theta(\Psi_i(m_i)) = 0.$$

Hence $\Psi_i(m_i) = 0$. The regularity assumption and Lemma 4.3 give $m_i = 0$. Cancellation of $s \otimes t^{\mathrm{op}}$ gives $k_i = 0$, and therefore $T_i = 0$. Remark 2.8 now gives $\Delta(X) = d$. □

Lemma 4.5. *Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$, and suppose that $X_i - \lambda$ is regular for every $1 \le i \le d$ and every $\lambda \in \mathbb{C}$. Let $(L_1, \dots, L_d)$ be a liberation dual system for X. If $k_1, \dots, k_d \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ satisfy*

$$h := \sum_{i=1}^{d} \Psi_i(k_i) \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}},$$

then, for every i,

$$k_i \in H \otimes_{\mathbb{C}} H, \qquad \overline{\Phi_i(h)} = \Theta(\Psi_i(k_i)).$$

Proof. Fix i, write $h = \sum_r c_r \otimes d_r^{\mathrm{op}}$ with $c_r, d_r \in \mathcal{M}$, and put $S_i := \Phi_i(h)$ on $\mathcal{M}\Omega$. The adjoint-domain condition in Definition 1.6 gives

$$\mathcal{M}\Omega \subseteq \operatorname{Dom} S_i^*, \qquad S_i^* \eta = \sum_r d_r^* L_i^*(c_r^* \eta) \quad (\eta \in \mathcal{M}\Omega).$$

Thus S_i is closable. By Lemma 3.1, choose regular $s, t \in \mathcal{M}$ such that

$$m_j := (s \otimes t^{\mathrm{op}})k_j \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}} \qquad (1 \le j \le d).$$

Applying Φ_i to $\sum_j \Psi_j(m_j) = (s \otimes t^{\mathrm{op}})h$ and using equation (17) gives

$$sS_i t = \Theta(\Psi_i(m_i)) \quad \text{on } \mathcal{M}\Omega. \tag{18}$$

The right-hand side has finite rank and s is injective, so the restriction of S_i to $t\mathcal{M}\Omega$ has finite rank. Since t has dense range, $t\mathcal{M}\Omega$ is dense in H. This restriction is closable, and Lemma 4.2 makes its closure a bounded finite-rank operator on H. The operator $\overline{S_i}$ extends that closure and therefore equals it. Set $h_i := \Theta^{-1}(\overline{S_i}) \in H \otimes_{\mathbb{C}} H$. Extending equation (18) by boundedness and density, and using the outer bimodule actions of Θ, we obtain

$$(s \otimes t^{\mathrm{op}})h_i = \Psi_i(m_i) = (s \otimes t^{\mathrm{op}})\Psi_i(k_i).$$

Cancellation of the regular outside factors gives $\Psi_i(k_i) = h_i \in H \otimes_{\mathbb{C}} H$. Since every $X_i - \lambda$ is regular, Lemma 4.3 gives $k_i \in H \otimes_{\mathbb{C}} H$. □

Proof of Theorem 1.7. By Proposition 4.4, $\Delta(X) = d$. We prove (1) $\Longrightarrow$ (4) $\Longrightarrow$ (3) $\Longrightarrow$ (2) $\Longrightarrow$ (1).

(1) $\Longrightarrow$ (4). Fix any fraction pair $Q = (s, a, b, t)$ for u. By Theorem 1.4(3), there are tensors $\ell_i \in L^0(\mathcal{M}) \otimes_{\mathbb{C}} L^0(\mathcal{M})^{\mathrm{op}}$ satisfying equation (5). Set

$$h := a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}} = \sum_i \Psi_i(\ell_i).$$

Lemma 4.5 gives $\ell_i \in H \otimes_{\mathbb{C}} H$ and $\overline{\Phi_i(h)} = \Theta(\Psi_i(\ell_i))$. By definition, $\Phi_i(h) = -\Gamma[Q, L_i]|_{\mathcal{M}\Omega}$. The common adjoint-domain condition makes $\Gamma[Q, L_i]$ closable by Lemma 5.1. Its closure extends

the bounded, everywhere-defined operator $-\overline{\Phi_i(h)}$ and therefore equals it. With $T_i := -\Theta(\ell_i) \in \mathcal{F}(H)$, equation (14) now gives

$$\overline{\Gamma[Q, L_i]} = -\Theta(\Psi_i(\ell_i)) = [\rho(X_i), T_i].$$

Thus assertion (4) holds.

(4) $\implies$ (3). Fix any fraction pair $Q = (s, a, b, t)$ for u and choose $T_i \in \mathcal{F}(H)$ such that $[\rho(X_i), T_i] = \overline{\Gamma[Q, L_i]}$ for every i. By equation (7), on $\mathcal{M}\Omega$ we have

$$\begin{aligned}\sum_i [\rho(X_i), T_i] &= \sum_i \overline{\Gamma[Q, L_i]} \\ &= s\Big(\sum_i L_i\Big)b - a\Big(\sum_i L_i\Big)t \\ &= sP_\Omega b - aP_\Omega t = \Gamma[Q, P_\Omega].\end{aligned}$$

Both sides are bounded, so density extends this identity to H, proving assertion (3).

(3) $\implies$ (2). Every affiliated operator admits a fraction pair, so this is immediate.

(2) $\implies$ (1). This is the implication (5) $\implies$ (1) of Theorem 1.4.

For each fixed fraction pair, uniqueness of the tuple $(T_1, \dots, T_d)$ follows from Theorem 1.4. □

5. Dual system criterion

We prove Theorem 1.9 by applying the liberation dual system criterion of Theorem 1.7. We first prove that the existence of a finite-rank closure and its rank are independent of the fraction pair, and identify finite-rank solutions of equation (6) with the closures of the corresponding fraction commutators. We then construct a normalized liberation dual system directly from the dual operators and apply assertion (4) of Theorem 1.7.

The following lemma supplies the domain facts needed for the fraction commutators.

Lemma 5.1. *Let $D_1, \dots, D_d$ be densely defined linear operators on H such that*

$$\mathcal{M}\Omega \subseteq \operatorname{Dom} D_i \cap \operatorname{Dom} D_i^*$$

for $1 \le i \le d$. For every $u \in L^0(\mathcal{M})$, every fraction pair Q of u, and every i, the operator $\Gamma[Q, D_i]$ is closable and densely defined.

Proof. Fix i and write $Q = (s, a, b, t)$. Since $b, t \in \mathcal{M}$ preserve $\mathcal{M}\Omega$, the domain of $\Gamma[Q, D_i]$ contains $\mathcal{M}\Omega$. For $\eta \in \mathcal{M}\Omega$ and $\xi \in \operatorname{Dom}\Gamma[Q, D_i]$, the hypothesis of the lemma gives

$$\langle \eta, \Gamma[Q, D_i]\xi\rangle = \langle b^* D_i^*(s^*\eta) - t^* D_i^*(a^*\eta), \xi\rangle\,. \tag{19}$$

Thus $\mathcal{M}\Omega \subseteq \operatorname{Dom}(\Gamma[Q, D_i])^*$, so $\Gamma[Q, D_i]$ is closable. □

The next lemma allows us to pass from one fraction pair to every fraction pair before applying Theorem 1.7.

Lemma 5.2. *Let D be a densely defined linear operator on H such that $\mathcal{M}\Omega \subseteq \operatorname{Dom} D \cap \operatorname{Dom} D^*$. For $u \in L^0(\mathcal{M})$ and any two fraction pairs Q, Q' for u,*

$$\overline{\Gamma[Q, D]} \in \mathcal{F}(H) \quad \Longleftrightarrow \quad \overline{\Gamma[Q', D]} \in \mathcal{F}(H).$$

When these conditions hold, the two closures have the same rank.

Proof. Write $Q = (s, a, b, t)$ and $Q' = (s', a', b', t')$. Lemma 2.6 gives regular $x, y, p, q \in \mathcal{M}$ such that $xs = ys'$ and $tp = t'q$. Since $xa = ya'$ and $bp = b'q$, we have on $\mathcal{M}\Omega$

$$x\Gamma[Q, D]p = y\Gamma[Q', D]q. \tag{20}$$

Suppose that $\overline{\Gamma[Q,D]} \in \mathcal{F}(H)$. The left-hand side of equation (20) has finite rank, and y is injective. Thus the restriction of $\Gamma[Q',D]$ to the dense subspace $q\mathcal{M}\Omega$ has finite rank. It is closable by Lemma 5.1, so Lemma 4.2 makes its closure a bounded finite-rank operator on all of H. The closure of $\Gamma[Q',D]$ extends this operator and hence equals it. Interchanging Q and Q' proves the equivalence.

Both closures are now bounded, so equation (20) extends to

$$x\overline{\Gamma[Q,D]}p = y\overline{\Gamma[Q',D]}q \qquad \text{on } H.$$

The left factors x, y are injective, while p, q have dense range. These multiplications preserve the rank of a bounded finite-rank operator, proving the rank equality. □

We next identify any finite-rank solution of equation (6) with the fraction commutators of a dual system.

Lemma 5.3. *Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$, and let $(D_1, \dots, D_d)$ be a dual system for X. Let Q be a fraction pair for $u \in L^0(\mathcal{M})$. If $T_1, \dots, T_d \in \mathcal{F}(H)$ satisfy*

$$\sum_j [\rho(X_j), T_j] = \Gamma[Q, P_\Omega],$$

then $\overline{\Gamma[Q,D_i]} = T_i$ for every i.

Proof. Write $Q = (s, a, b, t)$ and put $\ell_j := -\Theta^{-1}(T_j) \in H \otimes_{\mathbb{C}} H$. By Lemma 3.3,

$$\sum_j \Psi_j(\ell_j) = a \otimes t^{\mathrm{op}} - s \otimes b^{\mathrm{op}}.$$

Lemma 3.1 gives regular $s', t' \in \mathcal{M}$ such that $\ell_j' := (s' \otimes (t')^{\mathrm{op}})\ell_j \in \mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}}$ for every j. Equation (16) gives

$$\sum_j \Psi_j(\ell_j') = s'a \otimes (tt')^{\mathrm{op}} - s's \otimes (bt')^{\mathrm{op}}.$$

For fixed i and $\xi \in \mathcal{M}\Omega$, apply to this equality the linear map

$$\mathcal{M} \otimes_{\mathbb{C}} \mathcal{M}^{\mathrm{op}} \longrightarrow H, \qquad c \otimes e^{\mathrm{op}} \longmapsto cD_i(e\xi).$$

All inputs of D_i lie in $\mathcal{M}\Omega$, and the dual relations give

$$cX_jD_i(e\xi) - cD_i(X_je\xi) = \delta_{ij}cP_\Omega e\xi.$$

Using equation (13), we obtain

$$-s'\Gamma[Q,D_i]t'\xi = \Theta(\ell_i')\xi = -s'T_it'\xi.$$

Since s' is injective, $\Gamma[Q,D_i]$ agrees with T_i on the dense subspace $t'\mathcal{M}\Omega$. Lemma 5.1 gives closability of $\Gamma[Q,D_i]$. Its closure therefore extends the bounded operator T_i, whose domain is all of H, and hence equals T_i. □

For $N, S \in \mathcal{B}(H)$ with N normal, the classical Fuglede theorem [Fug50, Theorem I, p. 35] gives $SN = NS \implies SN^* = N^*S$. The following analogue for bounded linear maps from $\mathcal{M}$ to H will be used in Lemma 5.5.

Lemma 5.4 (see [CS77, Proposition 1])**.** *Let $x \in \mathcal{M}$ be normal, and let $T : \mathcal{M} \to H$ be a bounded linear map. If $T(xz) = xT(z)$ for every $z \in \mathcal{M}$, then $T(x^*z) = x^*T(z)$ for every $z \in \mathcal{M}$.*

Lemma 5.5. *Let $T : \mathcal{M}\Omega \to H$ be a closable linear operator. Then $z \mapsto T(z\Omega)$ is bounded from $\mathcal{M}$ to H. If $x_1, \dots, x_m \in \mathcal{M}$ are normal and $T(x_j\xi) = x_jT\xi$ for every j and $\xi \in \mathcal{M}\Omega$, then*

$$T(a\xi) = aT\xi \qquad (a \in W^*(x_1, \dots, x_m),\ \xi \in \mathcal{M}\Omega).$$

Proof. Identify $\mathcal{M}\Omega$ with $\mathcal{M}$. The graph of $T : \mathcal{M} \to H$ is norm closed because the Hilbert-space closure $\overline{T}$ is closed and agrees with T on $\mathcal{M}\Omega$. The closed graph theorem therefore gives the asserted bound. Lemma 5.4 gives $T(x_j^*\xi) = x_j^* T\xi$. Products and linear combinations give the same relation for the unital $*$-algebra generated by $x_1, \dots, x_m$. For $a \in W^*(x_1, \dots, x_m)$, Kaplansky density [AP18, Theorem 2.3.1] gives a uniformly bounded net a_λ in this algebra converging strongly to a. For fixed $\xi \in \mathcal{M}\Omega$,

$$a_\lambda \xi \longrightarrow a\xi, \qquad T(a_\lambda \xi) = a_\lambda T\xi \longrightarrow aT\xi.$$

Closedness of $\overline{T}$ and $a\xi \in \mathcal{M}\Omega$ give $T(a\xi) = aT\xi$. □

We now construct a normalized liberation dual system from a dual system for normal generators and compare their fraction commutators.

Lemma 5.6. *Let $X_1, \dots, X_d \in \mathcal{M}$ be normal operators generating $\mathcal{M}$, and let $(D_1, \dots, D_d)$ be a dual system for X. Then every $X_i - \lambda$, $\lambda \in \mathbb{C}$, is regular. Define on $\mathcal{M}\Omega$*

$$K_i := [\rho(X_i), D_i|_{\mathcal{M}\Omega}], \qquad C := \sum_j K_j - P_\Omega, \qquad L_i := K_i - \frac{1}{d} C. \tag{21}$$

The tuple $(L_1, \dots, L_d)$ is a normalized liberation dual system for X. For every fraction pair Q and every i, if $\overline{\Gamma[Q, D_i]}$ is bounded on H, then

$$\overline{\Gamma[Q, L_i]} = [\rho(X_i), \overline{\Gamma[Q, D_i]}]. \tag{22}$$

In particular, if $\overline{\Gamma[Q, D_i]} \in \mathcal{F}(H)$ for every i, then

$$\sum_{i=1}^{d} [\rho(X_i), \overline{\Gamma[Q, D_i]}] = \Gamma[Q, P_\Omega]. \tag{23}$$

Proof. For a spectral projection $e := 1_{\{\lambda\}}(X_i)$, the dual relation gives $eP_\Omega e = e[X_i, D_i]e = 0$ on $\mathcal{M}\Omega$. Density and faithfulness of the trace give $e = 0$. Normality then implies that $X_i - \lambda$ is regular.

Since $\rho(X_i)^* = \rho(X_i^*)$ preserves $\mathcal{M}\Omega$, the adjoint-domain hypothesis gives

$$\mathcal{M}\Omega \subseteq \operatorname{Dom} K_i^*, \qquad K_i^* \eta = D_i^* \rho(X_i^*)\eta - \rho(X_i^*) D_i^* \eta \quad (\eta \in \mathcal{M}\Omega).$$

Consequently, $\mathcal{M}\Omega$ is contained in the adjoint domains of C and each L_i as well. Left and right multipliers commute, and traciality gives $[\rho(X_i), P_\Omega] = [X_i, P_\Omega]$ on H. Jacobi's identity therefore gives on $\mathcal{M}\Omega$

$$[X_j, K_i] = [\rho(X_i), [X_j, D_i]] = \delta_{ij}[X_i, P_\Omega], \qquad [X_j, C] = 0.$$

It follows that $[X_j, L_i] = \delta_{ij}[X_i, P_\Omega]$ and $\sum_i L_i = P_\Omega$ on $\mathcal{M}\Omega$. Thus $(L_1, \dots, L_d)$ is a normalized liberation dual system.

The operator C is closable, so Lemma 5.5 gives $C(a\xi) = aC\xi$ for all $a \in \mathcal{M}$ and $\xi \in \mathcal{M}\Omega$. For $Q = (s, a, b, t)$, this implies

$$\Gamma[Q, C]\xi = (sb - at)C\xi = 0 \qquad (\xi \in \mathcal{M}\Omega).$$

Right multipliers commute with the left multipliers in Q, and hence

$$\Gamma[Q, L_i]\xi = \Gamma[Q, K_i]\xi = \rho(X_i)\Gamma[Q, D_i]\xi - \Gamma[Q, D_i]\rho(X_i)\xi \qquad (\xi \in \mathcal{M}\Omega). \tag{24}$$

If $\overline{\Gamma[Q, D_i]}$ is bounded on H, the right-hand side extends to $[\rho(X_i), \overline{\Gamma[Q, D_i]}] \in \mathcal{B}(H)$. By Lemma 5.1, $\Gamma[Q, L_i]$ is closable on its natural domain. Its closure extends this bounded operator on all of H, proving equation (22). Finally, summing equation (22) on $\mathcal{M}\Omega$ and using $\sum_i L_i = P_\Omega$ gives equation (23) by density. □

Proof of Theorem 1.9. Let $(L_1,\dots,L_d)$ be the normalized liberation dual system constructed in Lemma 5.6. That lemma also gives regularity of every scalar shift, so Theorem 1.7 applies. Fix $u\in L^0(\mathcal{M})$ and a fraction pair Q for u.

(1) $\Longrightarrow$ (3). By Theorem 1.7(4), there are $T_1,\dots,T_d\in\mathcal{F}(H)$ such that

$$\overline{\Gamma[Q,L_i]}=[\rho(X_i),T_i]\qquad(1\le i\le d).$$

Summing on $\mathcal{M}\Omega$ and using $\sum_i L_i=P_\Omega$, then extending by boundedness, gives

$$\sum_i[\rho(X_i),T_i]=\Gamma[Q,P_\Omega].$$

Lemma 5.3 gives $\overline{\Gamma[Q,D_i]}=T_i$ for every i, proving assertion (3).

(3) $\Longrightarrow$ (1). By Lemma 5.2, $\overline{\Gamma[Q',D_i]}\in\mathcal{F}(H)$ for every fraction pair Q' of u and every i. Equation (22) therefore gives

$$\overline{\Gamma[Q',L_i]}=[\rho(X_i),\overline{\Gamma[Q',D_i]}]\in[\rho(X_i),\mathcal{F}(H)].$$

Thus assertion (4) of Theorem 1.7 holds, and $u\in\operatorname{Div}_{L^0(\mathcal{M})}\mathcal{A}_X$.

(2) $\Longleftrightarrow$ (3). Lemma 5.1 gives closability, and Lemma 4.2 gives the forward implication. The reverse implication follows by restriction. The rank independence is Lemma 5.2. □

6. Applications to liberation gradients and conjugate variables

We first apply Theorem 1.7 to normal generators with L^2 liberation gradients and nonatomic spectral distributions. We then obtain the criterion for generators admitting conjugate variables as a corollary.

6.1. Liberation gradients.

We use the L^2 formulation of Voiculescu's liberation gradients in [Voi99, 5.3–5.4 and Proposition 6.1].

Definition 6.1. Let $X=(X_1,\dots,X_d)\in\mathcal{M}^d$ generate $\mathcal{M}$, and set

$$A_i:=W^*(X_i),\qquad B_i:=W^*(X_j:j\neq i),\qquad \mathcal{P}_i:=\operatorname{Alg}(A_i,B_i)\quad(1\le i\le d),$$

where $B_1:=\mathbb{C}1$ if $d=1$. We say that X admits L^2 *liberation gradients* if the derivations

$$\gamma_i:\mathcal{P}_i\subset H\longrightarrow\mathcal{P}_i\otimes_{\mathbb{C}}\mathcal{P}_i,\qquad \gamma_i(a):=a\otimes1-1\otimes a\quad(a\in A_i),\qquad \gamma_i(b):=0\quad(b\in B_i),$$

are well-defined and $1\otimes1\in\operatorname{Dom}\gamma_i^*$ for every i. Here γ_i^* is the Hilbert space adjoint for the dense domain $\mathcal{P}_i\Omega$ and the target $H\bar{\otimes}H$. The liberation gradients $j_i:=j(A_i:B_i):=-\gamma_i^*(1\otimes1)\in H$ satisfy

$$j_i^*=-j_i,\qquad \tau(j_ip)=(\tau\otimes\tau)(\gamma_ip)\quad(p\in\mathcal{P}_i).$$

This condition is equivalent to finiteness of the liberation Fisher information $\varphi^*(A_1;\dots;A_d):=\sum_i\|j_i\|_2^2$ in [Voi99, Definition 9.1].

The following construction supplies the operators needed for Theorem 1.7.

Lemma 6.2. *If X admits L^2 liberation gradients in the sense of Definition 6.1, then X admits a normalized liberation dual system.*

Proof. Fix i, and define on $\mathcal{P}_i$

$$\delta_i:=\mathrm{i}\gamma_i,\qquad C_i^\circ(p\Omega):=(\mathrm{id}\otimes\tau)\delta_ip,\qquad \eta:=\mathrm{i}j_i.$$

Then $\delta_i^*(1\otimes1)=\eta$. The prescriptions on A_i,B_i and the product rule show that δ_i is real: $\delta_i(p^*)=(\delta_ip)^\dagger$, where $(u\otimes v)^\dagger:=v^*\otimes u^*$. By [Mai15, Lemmas 4.6–4.7], $\eta=\eta^*$, the algebraic tensor product $\mathcal{P}_i\otimes_{\mathbb{C}}\mathcal{P}_i$ is contained in $\operatorname{Dom}\delta_i^*$, and

$$\langle C_i^\circ p,q\rangle=\langle p,q\eta-C_i^\circ q\rangle\qquad(p,q\in\mathcal{P}_i).$$

In particular, both δ_i and C_i° are closable. We adapt the argument in [Dab10, proof of Lemma 12]. For $\delta_i p = \sum_r u_r \otimes v_r$, the prescriptions on A_i, B_i and the product rule give

$$(\delta_i \otimes \mathrm{id})\delta_i p - (\mathrm{id} \otimes \delta_i)\delta_i p = \mathrm{i} \sum_r u_r \otimes 1 \otimes v_r.$$

Indeed, the mixed product-rule terms cancel, and the two sides agree on A_i and B_i. Since $C_i^\circ(\mathcal{P}_i) \subseteq \mathcal{P}_i$, applying $\mathrm{id} \otimes \tau \otimes \tau$ and using $\delta_i^*(1 \otimes 1) = \eta$ gives

$$(C_i^\circ)^2 p = \sum_r u_r \tau(v_r \eta) + \mathrm{i} C_i^\circ p.$$

The preceding integration by parts identity and the expansion of $\delta_i(p^* p)$ yield

$$\|C_i^\circ p\|_2^2 = 2\,\mathrm{Re}\langle C_i^\circ p, p\eta\rangle - \langle \delta_i(p^* p), 1 \otimes \eta\rangle - \langle \mathrm{i} C_i^\circ p, p\rangle. \tag{25}$$

For $a = a^* \in \mathcal{P}_i$ with $\|a\| < 1$, set $v := a + \mathrm{i}(1 - a^2)^{1/2}$. The polynomial partial sums v_n of its power series converge in operator norm. The series for $\delta_i v_n$ and $\delta_i(v_n^*)$ converge in $H \bar{\otimes} H$, since $\|\delta_i(a^n)\|_2 \le n\|a\|^{n-1}\|\delta_i a\|_2$. The product rule shows that $\delta_i(v_n^* v_n)$ converges, while $v_n^* v_n \to 1$ in operator norm. Closedness of $\overline{\delta_i}$ therefore gives $\delta_i(v_n^* v_n) \to 0$. Taking real parts and passing to the limit in (25), with $\|v\|_2 = 1$, gives

$$\lim_n \|C_i^\circ v_n\|_2 \le 2\|\eta\|_2 + 1.$$

Apply the same argument to v^*; averaging via $a = (v + v^*)/2$, rescaling and splitting real and imaginary parts give, for some $K_i > 0$,

$$\|C_i^\circ p\|_2 \le K_i \|p\| \qquad (p \in \mathcal{P}_i).$$

The integration by parts identity shows that $\mathcal{P}_i\Omega \subseteq \mathrm{Dom}(C_i^\circ)^*$ and

$$(C_i^\circ)^*(p\Omega) = p\eta - C_i^\circ(p\Omega), \qquad \|(C_i^\circ)^*(p\Omega)\|_2 \le (K_i + \|\eta\|_2)\|p\| \quad (p \in \mathcal{P}_i).$$

Put $C_i := \overline{C_i^\circ}$. For $x \in \mathcal{M}$, Kaplansky density [AP18, Theorem 2.3.1] gives a uniformly bounded net $p_\lambda \in \mathcal{P}_i$ with $p_\lambda\Omega \to x\Omega$. The estimate for $C_i^\circ p_\lambda$ gives a weakly convergent subnet in H. The graph of C_i is a closed linear subspace of $H \oplus H$ and hence is weakly closed, so $x\Omega \in \mathrm{Dom}\, C_i$. Applying the same argument to the bound for $(C_i^\circ)^* p_\lambda$ gives $x\Omega \in \mathrm{Dom}\, C_i^*$. Thus

$$\mathcal{M}\Omega \subseteq \mathrm{Dom}\, C_i \cap \mathrm{Dom}\, C_i^*.$$

On $\mathcal{P}_i\Omega$, the product rule and the values of δ_i give

$$[C_i^\circ, X_j] = \mathrm{i}\delta_{ij}[X_i, P_\Omega].$$

Since $\mathcal{P}_i\Omega$ is a core for C_i and the right-hand side is bounded, approximation in the graph norm extends this identity to $\mathrm{Dom}\, C_i$, with $X_j\,\mathrm{Dom}\, C_i \subseteq \mathrm{Dom}\, C_i$. In particular it holds on $\mathcal{M}\Omega$. Set $L_i := \mathrm{i}C_i$. Then both $\mathrm{Dom}\, L_i$ and $\mathrm{Dom}\, L_i^*$ contain $\mathcal{M}\Omega$, and

$$[X_j, L_i]|_{\mathcal{M}\Omega} = \delta_{ij}[X_i, P_\Omega]|_{\mathcal{M}\Omega}.$$

This is a liberation dual system. Lemma 4.1 supplies a normalized system. □

For normal generators, nonatomicity supplies the remaining regularity hypothesis of Theorem 1.7.

Proposition 6.3. *Let $X_1, \ldots, X_d \in \mathcal{M}$ be normal operators generating $\mathcal{M}$, each with nonatomic spectral distribution. Suppose that X admits L^2 liberation gradients in the sense of Definition 6.1. Then $\Delta(X) = d$, and X admits a normalized liberation dual system. For any such system, all the*

conclusions of Theorem 1.7 hold. In particular, for every $u \in L^0(\mathcal{M})$ and every fraction pair Q for u, we have $u \in \mathrm{Div}_{L^0(\mathcal{M})} \mathcal{A}_X$ if and only if there exist $T_1, \dots, T_d \in \mathcal{F}(H)$ such that

$$\sum_i [\rho(X_i), T_i] = \Gamma[Q, P_\Omega].$$

For each fixed Q, the tuple $(T_1, \dots, T_d)$ is unique whenever it exists.

Proof. Lemma 6.2 gives a normalized liberation dual system. Normality and nonatomicity imply that every $X_i - \lambda$, $\lambda \in \mathbb{C}$, is regular. Theorem 1.7 gives the criterion and uniqueness, and Proposition 4.4 gives $\Delta(X) = d$. □

6.2. **Conjugate variables.** We now deduce the ordinary conjugate-variable case. We use the following formulation, as in [ACSY24, Proposition 3.5].

Definition 6.4. Let $X = (X_1, \dots, X_d) \in \mathcal{M}^d$ generate $\mathcal{M}$, and write

$$\mathcal{A}_X^* := \mathbb{C}\langle X_1, \dots, X_d, X_1^*, \dots, X_d^* \rangle.$$

We say that X admits *conjugate variables* if there exist non-commutative derivations

$$\partial_i : \mathcal{A}_X^* \subset H \longrightarrow H \bar{\otimes} H \qquad (1 \le i \le d)$$

such that

$$\partial_i(X_j) = \delta_{ij} 1 \otimes 1, \qquad 1 \otimes 1 \in \mathrm{Dom}\, \partial_i^* \quad (1 \le i, j \le d). \tag{26}$$

The vectors $\xi_i := \partial_i^*(1 \otimes 1) \in H$ are called conjugate variables for X.

The following commutator equation will identify the derivatives of the adjoint generators. For a normal operator $x \in \mathcal{M}$ with nonatomic spectral distribution μ, define in $L^2(\mu) \bar{\otimes} L^2(\mu) \subseteq H \bar{\otimes} H$

$$v_x(z, w) := \begin{cases} (\bar{z} - \bar{w})/(z - w), & z \neq w, \\ 0, & z = w, \end{cases} \qquad K_x := \Theta(v_x) \in S_2(H). \tag{27}$$

Lemma 6.5. *Let $x \in \mathcal{M}$ be a normal operator with nonatomic spectral distribution μ. For every $c \in \mathbb{C}$, the equation*

$$[K, x] = c[P_\Omega, x^*]$$

has the unique solution $K = cK_x$ in $S_2(H)$.

Proof. The diagonal has $\mu \otimes \mu$ measure zero and $|v_x| = 1$ off it. Multiplication of its kernel gives

$$[K_x, x] = [P_\Omega, x^*].$$

If two solutions differ by $S \in S_2(H)$, then $[S, x] = 0$. The classical Fuglede theorem gives $[S, x^*] = 0$, so the positive compact operator S^*S commutes with x. If $S \neq 0$, a positive eigenvalue of S^*S has a nonzero finite-dimensional eigenspace invariant under x. The restriction of x to this space has an eigenvector, contrary to the absence of spectral atoms. Thus $S = 0$. □

We next verify the hypotheses of Proposition 6.3.

Lemma 6.6. *Let $X_1, \dots, X_d \in \mathcal{M}$ be normal operators generating $\mathcal{M}$. Suppose that X admits conjugate variables in the sense of Definition 6.4. Then each X_j has nonatomic spectral distribution, and X admits L^2 liberation gradients.*

Proof. Choose derivations $\partial_1, \dots, \partial_d$ as in Definition 6.4. By [ACSY24, Lemma 3.3], the algebraic tensor product $\mathcal{A}_X^* \otimes_{\mathbb{C}} \mathcal{A}_X^*$ is contained in $\mathrm{Dom}\, \partial_i^*$, so each ∂_i is closable. Fix i, $\lambda \in \mathbb{C}$ and $\varepsilon > 0$, and set

$$y := X_i - \lambda, \qquad e := 1_{\{\lambda\}}(X_i), \qquad r_\varepsilon := (y^* y + \varepsilon^2)^{-1}, \qquad f_\varepsilon := \varepsilon^2 y r_\varepsilon.$$

To check the domains, choose $R > \|y\|^2 + \varepsilon^2$ and put $q := 1 - R^{-1}(y^*y + \varepsilon^2) \in \mathcal{A}_X^*$. Then $\|q\| < 1$ and

$$r_\varepsilon = R^{-1} \sum_{n=0}^{\infty} q^n, \qquad \|\partial_i(q^n)\|_2 \le n\|q\|^{n-1}\|\partial_i q\|_2 \quad (n \ge 1).$$

The partial sums converge in operator norm and their images under ∂_i converge in $H \bar{\otimes} H$. Thus $r_\varepsilon \in \operatorname{Dom} \overline{\partial_i}$, with

$$\overline{\partial_i} r_\varepsilon = -r_\varepsilon \cdot (\partial_i(y^*) \cdot y + y^* \cdot \partial_i y) \cdot r_\varepsilon.$$

Multiplying the partial sums by $\varepsilon^2 y$ and using the product rule also gives $f_\varepsilon \in \operatorname{Dom} \overline{\partial_i}$. Since y is normal, $y r_\varepsilon y^* = 1 - \varepsilon^2 r_\varepsilon$, and hence

$$\begin{aligned} \overline{\partial_i} f_\varepsilon &= \varepsilon^4 r_\varepsilon \cdot \partial_i y \cdot r_\varepsilon - \varepsilon^2 y r_\varepsilon \cdot \partial_i(y^*) \cdot y r_\varepsilon \\ &= (\varepsilon^2 r_\varepsilon) \otimes (\varepsilon^2 r_\varepsilon) - (\varepsilon y r_\varepsilon) \cdot \partial_i(y^*) \cdot (\varepsilon y r_\varepsilon). \end{aligned}$$

Spectral calculus gives

$$\|\varepsilon^2 r_\varepsilon\| \le 1, \qquad \|\varepsilon y r_\varepsilon\| \le \tfrac{1}{2}, \qquad \varepsilon^2 r_\varepsilon \longrightarrow e, \qquad \varepsilon y r_\varepsilon \longrightarrow 0 \quad \text{strongly as } \varepsilon \downarrow 0.$$

Consequently, $\overline{\partial_i} f_\varepsilon \to e \otimes e$ in $H \bar{\otimes} H$. On the other hand, $\|f_\varepsilon\| \le \varepsilon/2$, so $f_\varepsilon \to 0$ in H. Closedness of $\overline{\partial_i}$ gives $e \otimes e = 0$, and faithfulness of τ gives $e = 0$. Since i and λ were arbitrary, every X_j has nonatomic spectral distribution. Fix i, and write $\xi_i := \partial_i^*(1 \otimes 1)$. The map Ψ_i acts on $H \bar{\otimes} H$ by the same formula. On $\mathcal{A}_X^*$, define $\gamma_i^\circ := \Psi_i \partial_i$. Differentiating $X_j X_j^* = X_j^* X_j$ and applying Lemma 6.5 gives $\partial_i X_j^* = \delta_{ij} v_{X_j}$, hence

$$\begin{aligned} \gamma_i^\circ(X_j) &= \delta_{ij}(X_j \otimes 1 - 1 \otimes X_j), \\ \gamma_i^\circ(X_j^*) &= \delta_{ij}(X_j^* \otimes 1 - 1 \otimes X_j^*). \end{aligned}$$

Since Ψ_i commutes with the outer bimodule actions, γ_i° is a derivation with algebraic tensor values, and these identities make it anti-real. The tensor-adjoint formula [Mai15, Lemma 4.3] gives

$$(\gamma_i^\circ)^*(1 \otimes 1) = [X_i^*, \xi_i].$$

Consequently, $j_i := [\xi_i, X_i^*] \in H$ is skewadjoint and satisfies

$$\tau(j_i p) = (\tau \otimes \tau)(\gamma_i^\circ p) \qquad (p \in \mathcal{A}_X^*).$$

The corresponding word-moment formula holds for the algebras $\mathbb{C}[X_i, X_i^*]$ and $\mathbb{C}\langle X_j, X_j^* : j \ne i\rangle$. By bounded Kaplansky approximation in each factor, it extends to $A_i = W^*(X_i)$ and $B_i = W^*(X_j : j \ne i)$, with the same j_i; see [Voi99, Remark 5.5(c)]. To check well-definedness, let r be a formal relation between A_i and B_i that evaluates to zero, and let $\widetilde{\gamma}_i(r)$ be its evaluated formal tensor derivative. Applying the word-moment formula to urv and using the product rule gives

$$(\tau \otimes \tau)(u \cdot \widetilde{\gamma}_i(r) \cdot v) = 0 \qquad (u, v \in \mathcal{P}_i).$$

The other product-rule terms contain the vanishing factor r. The faithful product-trace pairing therefore gives $\widetilde{\gamma}_i(r) = 0$, so the derivation descends to $\mathcal{P}_i = \operatorname{Alg}(A_i, B_i)$. The resulting L^2 moment identity gives $1 \otimes 1 \in \operatorname{Dom} \gamma_i^*$ and $\gamma_i^*(1 \otimes 1) = -j_i$. Thus X admits L^2 liberation gradients in the sense of Definition 6.1. □

Corollary 6.7. *Let $X_1, \dots, X_d \in \mathcal{M}$ be normal operators generating $\mathcal{M}$. Suppose that X admits conjugate variables in the sense of Definition 6.4. Then all the conclusions of Proposition 6.3 hold. In particular, $\Delta(X) = d$, and for any normalized liberation dual system for X, all the conclusions of Theorem 1.7 hold.*

Proof. Lemma 6.6 gives nonatomic spectral distributions and L^2 liberation gradients, so Proposition 6.3 applies. □

Remark 6.8. The existence of a liberation dual system does not imply the existence of conjugate variables, even for normal generators with nonatomic spectral distributions. Indeed, let

$$d\mu(x) := \frac{dx}{\pi\sqrt{1-x^2}} \quad (-1 < x < 1), \qquad \mathcal{M} := L^\infty((-1,1),\mu), \qquad \tau(f) := \int f\, d\mu,$$

and let $X(x) := x$. Then $X = X^*$ generates $\mathcal{M}$ and has nonatomic spectral distribution, so every $X - \lambda$, $\lambda \in \mathbb{C}$, is regular. Taking $L := P_\Omega$ gives a normalized liberation dual system, since $[X, L] = [X, P_\Omega]$ and L is bounded on H. The liberation derivation relative to scalars is $a \mapsto a \otimes 1 - 1 \otimes a$, whose trace pairing is zero, so $j(W^*(X) : \mathbb{C}) = 0$ and Proposition 6.3 applies.

Suppose that X admits a conjugate variable in the sense of Definition 6.4, and choose the corresponding derivation ∂. Since $\partial X = 1 \otimes_{\mathbb{C}} 1$, the product rule and the moments $\tau(X^{2r}) = 4^{-r}\binom{2r}{r}$ and $\tau(X^{2r+1}) = 0$ give

$$(\tau \otimes \tau)(\partial X^{2n+1}) = 4^{-n} \sum_{r=0}^{n} \binom{2r}{r}\binom{2n-2r}{n-r} = 1,$$

where the last equality follows by comparing coefficients in $(1-z)^{-1/2}(1-z)^{-1/2} = (1-z)^{-1}$. On the other hand, $\|X^{2n+1}\|_2 \to 0$ by dominated convergence. Thus $\xi := \partial^*(1 \otimes_{\mathbb{C}} 1) \in H$ would satisfy

$$1 = \left|(\tau \otimes \tau)(\partial X^{2n+1})\right| \le \|\xi\|_2 \, \|X^{2n+1}\|_2 \longrightarrow 0,$$

a contradiction.

7. Examples

A circular operator illustrates the distinction between the criteria in Theorems 1.7 and 1.9.

Remark 7.1. The normality assumption in Theorem 1.9 cannot be omitted. Let l_j, r_j, $j = 1, 2$, be the left and right creation operators on a full Fock space with the vacuum vector Ω. Their adjoints l_j^*, r_j^* are the left and right annihilation operators, respectively. The operators $S_j := l_j + l_j^*$ form a standard free semicircular pair; see [Voi98, 5.13]. Set

$$c := \frac{S_1 + \mathrm{i}S_2}{\sqrt{2}}, \qquad \mathcal{M} := W^*(c), \qquad D := -\frac{r_1^* - \mathrm{i}r_2^*}{\sqrt{2}} \in \mathcal{B}(H).$$

The identities $[r_i^*, S_j] = \delta_{ij} P_\Omega$ give $[c, D] = P_\Omega$ and $[D, c^*] = 0$. Thus, for every fraction pair $Q = (s, a, b, t)$ for c^*,

$$\Gamma[Q, D] = s[D, c^*]t = 0 \in \mathcal{F}(H).$$

However, $c^* \notin \mathrm{Div}_{L^0(\mathcal{M})} \mathbb{C}[c]$: every element of this division closure commutes with c, whereas $[c, c^*] = \mathrm{i}[S_2, S_1] \neq 0$.

The bounded dual D also gives $\Delta(c) = 1$ by [MSY23, Proposition 6.2], so Theorem 1.4 still applies to c. Since c^* is not rational in c, none of its equivalent conditions holds for c^*. In particular, the equation in Theorem 1.4(5) has no finite-rank solution for any fraction pair for c^*. For every fraction pair $Q = (s, a, b, t)$ for c^*, $\Gamma[Q, D] = 0$ and thus fails to solve this equation:

$$[\rho(c), \Gamma[Q, D]] = 0 \neq s[P_\Omega, c^*]t = \Gamma[Q, P_\Omega].$$

Here $[P_\Omega, c^*] \neq 0$ since c^* is centered and nonzero. Regularity makes s injective and t have dense range, so $s[P_\Omega, c^*]t \neq 0$.

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